1. Confusion Matrix

We will calculate True Positives (TP), False Positives (FP), and False Negatives (FN) for each class.

- For `budget_planning`:
  - TP (Predicted = BP, Actual = BP): 14
  - FP (Predicted = BP, Actual ≠ BP): 4 (Rows: 3, 5, 10, 26)
  - FN (Predicted ≠ BP, Actual = BP): 1 (Row: 19)
- For `general_qa`:
  - TP (Predicted = GQA, Actual = GQA): 11
  - FP (Predicted = GQA, Actual ≠ GQA): 1 (Row: 19)
  - FN (Predicted ≠ GQA, Actual = GQA): 4 (Rows: 3, 5, 10, 26)
- For `risk_profiling`:
  - TP (Predicted = RP, Actual = RP): 5
  - FP (Predicted = RP, Actual ≠ RP): 0
  - FN (Predicted ≠ RP, Actual = RP): 1 (Row: 14)
- For `savings_optimization`:
  - TP (Predicted = SO, Actual = SO): 2
  - FP (Predicted = SO, Actual ≠ SO): 1 (Row: 14)
  - FN (Predicted ≠ SO, Actual = SO): 0

2. Overall Accuracy

Accuracy is the proportion of total correct predictions.

Correct Predictions: We can count the rows where Predicted == Actual.
From the file, the number of correct predictions is 30.

Total Predictions: 35

Accuracy = (Number of Correct Predictions) / (Total Predictions) = 30 / 35
Accuracy = 0.8571 or 85.71%

3. Precision, Recall, and F1-Score by Class

The formulas are:

- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)

- F1-Score = 2 * (Precision * Recall) / (Precision + Recall)

Let's calculate for each class:

`budget_planning`:

- Precision = 14 / (14 + 4) = 14 / 18 = 0.7778
- Recall = 14 / (14 + 1) = 14 / 15 = 0.9333
- F1-Score = 2 * (0.7778 * 0.9333) / (0.7778 + 0.9333) = 2 * 0.7259 / 1.7111 ≈ 0.8485

`general_qa`:

- Precision = 11 / (11 + 1) = 11 / 12 = 0.9167
- Recall = 11 / (11 + 4) = 11 / 15 = 0.7333
- F1-Score = 2 * (0.9167 * 0.7333) / (0.9167 + 0.7333) = 2 * 0.6722 / 1.65 ≈ 0.8148

`risk_profiling`:

- Precision = 5 / (5 + 0) = 5 / 5 = 1.0000
- Recall = 5 / (5 + 1) = 5 / 6 = 0.8333
- F1-Score = 2 * (1.0000 * 0.8333) / (1.0000 + 0.8333) = 2 * 0.8333 / 1.8333 ≈ 0.9091

`savings_optimization`:

- Precision = 2 / (2 + 1) = 2 / 3 = 0.6667
- Recall = 2 / (2 + 0) = 2 / 2 = 1.0000
- F1-Score = 2 * (0.6667 * 1.0000) / (0.6667 + 1.0000) = 2 * 0.6667 / 1.6667 = 0.8000

4. Summary

Metric	Overall	budget_planning	general_qa	risk_profiling	savings_optimization
Precision	-	77.78%	91.67%	100.00%	66.67%
Recall	-	93.33%	73.33%	83.33%	100.00%
F1-Score	-	84.85%	81.48%	90.91%	80.00%

Accuracy	85.71%

Analysis:

- The model has high overall accuracy (85.71%).
- `risk_profiling` has perfect precision, meaning whenever it predicts this class, it is always correct.
- `savings_optimization` has perfect recall, meaning it successfully identified all actual instances of this class (though it sometimes mistakes other classes for it).
- The main errors come from misclassifying general commands (like "conda activate karan") and simple income statements as `budget_planning`, and from confusing `risk_profiling` with `savings_optimization`.